Relational-Geometric Learning over the Software Development Lifecycle

Program report — data, pipelines, methods, hypotheses, experiments, results, and next steps

Relational SDLC AI Lab · public research repository · compiled 2026-06-23 · state through wave R19

At a glance	
Thesis	Embeddings (and later SLMs) trained over **verifiable SDLC relations** generalize across repositories better than generic text similarity.
Headline win	Relation-loss **LoRA inside the representation** beats the frozen embedder cross-repo, and the gain **grows with data**: ΔR@1 +0.07 (pilot) → +0.080 (55 repos) → +0.114 (78-repo dense Tier-2) → **+0.142** with same-repo **hard negatives** (R18/R19, first wave trained on the Apple M5 GPU).
Robustness	Positive on **all 5** held-out-repo splits (ΔR@1 +0.061±0.021); on the headline split both query- and repo-cluster **95% CIs exclude zero** (R17a). A thin parameter-free **graph lift** stacks on top (LoRA+graph R@1 0.690), robust across α and saturated at 1 hop.
Strongest negatives	Learned heads on *frozen* vectors overfit at pilot scale; a code-MLM base collapses; **more text hurts** (de-truncation −0.09 to −0.15 R@1).
Discipline	CPU-first, one change vs. the embedder-cosine control, de-referenced cross-repo splits, every run emits an experiment card, honest negatives kept.

Abstract

This report synthesizes the work of a public research lab studying *relational-geometric learning* over software-development-lifecycle (SDLC) records. The lab tests one claim: that learning over **verifiable relations** among artifacts — a pull request *fixes* an issue, a commit *modifies* a file, a test *covers* a symbol — yields retrieval that generalizes across repositories better than generic text similarity. We instantiate a typed artifact/relation graph, a ladder of relation operators (diagonal metric, low-rank two-tower, identity-initialized operator, a typed graph-aggregation lift, and a LoRA-fine-tuned encoder), and a frozen, leakage-guarded benchmark scored by Recall@K, MRR, and hard-negative accuracy. The arc is deliberately one-change-at-a-time against an *embedder-cosine* control. The settled findings: pretrained embeddings are the cross-repo substrate; the relational contribution must live **inside the representation** (LoRA), not in a head bolted onto frozen vectors; and the LoRA win is robust and grows with data density. We report the datasets, pipelines, methods, hypotheses, the full scope of experiments (waves R3–R16E), results, conclusions, and next steps.

1 Overview and Thesis

Modern software-engineering assistants are built largely on *text* retrieval: an embedding model maps an artifact to a vector and relevance is cosine similarity. That works when the answer restates the question, but SDLC artifacts are bound by **relations** that are often not surface-visible yet are *verifiable* — minable from closing keywords, diff contents, or coverage maps — which makes them an unusually trustworthy supervision signal.

Thesis. Embeddings (and later small language models) trained over verifiable SDLC relations produce more reliable software-engineering systems than models that rely on generic text similarity.

The first target is narrow on purpose: not an autonomous coding agent, but a **measurable relational retrieval layer** — issue → likely fixing PRs; diff → affected tests; failing log → likely files; PR → risk and missing-test candidates. This is MVP-1 of the program. The enduring deliverable is not any single model but a **frozen public benchmark with honest baselines and full provenance** that makes these questions falsifiable. Five standing goals govern the work: public research hygiene (G1), falsifiable retrieval gains (G2), dataset provenance (G3), relation-aware models (G4), and agentic readiness (G5).

2 Data and Datasets

Every dataset is a frozen snapshot of public, permissively-licensed GitHub repositories (plus one original synthetic fixture). Records are typed SDLC artifacts; edges are typed relations. Each record and edge carries full provenance — **source_url**, **retrieved_at**, **license**, a real **sha256** content hash, a transform lineage, an extraction **method**, and a **valid_from** timestamp used for temporal leakage control. JSON schemas for records, edges, benchmark queries, and the source/dataset/experiment cards live under **schemas/**.

2.1 Dataset tiers

Dataset	Repos	Records	fixes edges	Bench queries	q / repo	Role
datebox (synthetic)	—	original	controlled	94 held-out	—	Mechanism probe (CC0); timezone-bug worked example; runs offline from a clean checkout
pilot (P1)	20	2,087	356	356	~18	First real snapshot; issue→PR; the workhorse for operator ablations
scale (Tier-2 entry)	55	3,672	562	562	~10	Breadth check; bag-of-tokens + LoRA at larger repo count
tier2 (dense)	78	16,998	2,744	2,744	~35	Wide and ~3.5x denser per repo; the representation-learning regime

Table 1. Frozen dataset tiers. Tier-2 records are 2,282 issue + 14,716 pull_request; the snapshot is pruned to records referenced by an edge or query. Prior snapshots are never mutated (other experiments depend on them) and ids are namespaced and disjoint (pinned by a test).

Two further *derived* sets support specific waves: a **full-text** variant of the pilot that de-truncates issue/PR bodies (500 → 8000 chars) for the paired truncation control (R14), and a **deep-content** set of long file bodies for the diff → affected-test chunking study (R16A). The graph-learning track adds file-level structure to the pilot via a one-time live enrichment: **497 file + 239 test nodes** (736 total) and **1,356 *modifies* edges** over 18 repos, each edge observed directly from a diff with confidence 1.0.

2.2 Honesty by construction

- **Links are labels, not features.** Explicit references (**#N**, **owner/repo#N**, issue/PR URLs, commit SHAs) are scrubbed from inputs while kept as ground truth — a relation a regex can recover is a label source, never a test.
- **Frozen, cross-repo splits.** Train repositories are held disjoint from test repositories, so a win must generalize to unseen vocabularies and APIs; the default real-data split is temporal-by-commit-date. Splits are recorded in a dataset card.
- **Temporal leakage guard.** A query's optional **as_of** time excludes any candidate or positive that only becomes valid later; the validator exits non-zero on leakage.
- **Hard negatives are the point.** Each query carries near-miss negatives (wrong file in the same package, wrong test in the same suite, plausible-but-unrelated PR); without them Recall@K flatters every model.
- **Redistribution is metadata-only.** Provenance points back to public URLs; full source text is not redistributed in the public repo.

3 Pipelines

The pipeline turns public GitHub data into a provenance-bearing, leakage-guarded, frozen benchmark, and then replays cheaply on cached features. It is methodology-as-code, and every stage is reproducible from a clean checkout with numpy alone — no GPU and no network on the evaluation path.

Stage	What it does	Key property
1. Ingest	Map issue / PR / commit JSON and a PR's changed files into typed records and *fixes* / *modifies* edges (pure standard library).	Offline by default; live fetch is opt-in behind a flag + token, polite (rate-limit-aware, paced), recorded once then replayed.
2. Scrub	Remove explicit cross-references from input text; keep them as labels.	Makes the task semantic, not pointer-chasing.
3. Validate	Enforce JSON schemas, provenance completeness, referential integrity, and the temporal leakage guard over the whole data tree.	Exits non-zero on any error; 0 errors required to proceed.
4. Embed	Encode record text with a frozen small embedder (MiniLM-L6, 384-d) once; cache to .npz.	Heavy step cached and (for small tiers) committed, so eval is numpy-only.
5. Graph-enrich	Fetch fixing-PR changed files; add file/test nodes and *modifies* edges with provenance.	Unblocks the graph track and diff→test.
6. Benchmark + eval	Build frozen cross-repo queries with hard negatives; score Recall@K / MRR / hard-neg; apply the graph leakage guard (gold edge removed before aggregation).	Deterministic given a seed; one change at a time vs. the control.
7. Cards	Emit a dataset / experiment card per run; record the outcome in the §results ledger.	Exploratory results labelled as such; cards are the source of truth.

Table 2. The seven-stage pipeline. CI never runs the live builder; it validates the committed snapshot and re-runs the deterministic numpy ablation. An independent audit (R13) confirmed every numpy result reproduces the committed cards byte-for-byte.

4 Methods and Techniques

All systems are scored on the *same* candidate pools, so every comparison is attributable to one change. The operators form a ladder, each testing a different hypothesis about *where* the relational signal lives.

System / technique	What it is	Tests
Vanilla cosine	Plain cosine over bag-of-token vectors — the off-the-shelf text floor.	The similarity baseline.
Unsupervised IDF cosine	Cosine with corpus IDF token weights; no relation labels.	Is a gain just a frequency effect?
Diagonal relation metric	Non-negative per-token weights from a margin triplet loss over train-split *fixes* pairs (reweights shared dimensions only).	Is the relation a per-token reweighting?
Two-tower projection	Asymmetric low-rank query/document projections (margin triplet); can align *different* tokens across sides.	Can a learned cross-token operator over bag-of-tokens generalize?
Identity-init operator	A relation map M initialized at identity (starts exactly at embedder-cosine), regularized back, on *frozen* embeddings.	Can a safe refinement of frozen vectors help?
LoRA-tuned encoder	Low-rank adapters (r=8, α=16; ~0.48% of params) on MiniLM-L6 attention q/k/v, symmetric InfoNCE over train-repo pairs; pretrained weights untouched.	Does the relation loss *inside* the representation generalize cross-repo?
Typed graph lift	Training-free typed mean-aggregation: aug(v)=normalize(α·own + (1−α)·mean over relation-neighbours); swept over α and hops.	Does graph structure add signal beyond pairwise cosine?
Learned R-GCN	A 1-hop relational GCN, init at frozen features, InfoNCE.	Does a learned GNN beat parameter-free aggregation?

System / technique	What it is	Tests
Chunking (FirstP / MaxP)	Score by the lede chunk (FirstP) or the best chunk (MaxP) over a chunked body.	Where is the signal — front-loaded or deep?
SLM packer (MVP-2 seed)	Relation-conditioned subgraph retrieval packs context for a small LM (dry-run only).	Can a relation-packed subgraph drive an SLM?

Table 3. The method ladder. The bag-of-tokens and graph systems are numpy-only and deterministic; the embedder systems use a frozen MiniLM-L6 (22M params, CPU), embedded once and cached so evaluation needs no torch.

4.1 Objective and metrics

Operators are trained with a **topological margin loss** — a triplet objective $L\_topo = \Sigma [\gamma - s\_r(u,v^+) + s\_r(u,v^-)]_+$ — realized at scale as a symmetric InfoNCE / multiple-negatives contrastive loss with in-batch negatives. The relation score is s_r(u,v) = f_r(h_u, h_v): a diagonal W_r (per-token reweighting), a low-rank W_r = Wq·Wd (two-tower), or a translational map h_v ≈ M_r·h_u. Systems are judged by **Recall@K** (K = 1, 5, 10), **MRR**, and **hard-negative accuracy**. R@1 and MRR are the discriminating metrics; at the small candidate pools here R@5/R@10 are near-ceiling.

5 Hypotheses

The program is organized around six falsifiable questions, each with a control it must beat and a current verdict from the evidence.

#	Falsifiable hypothesis	Status
Q1	Fine-tuning a small embedder with the relation/contrastive loss (LoRA) beats frozen embedder-cosine cross-repo.	**Confirmed**, and grows with scale/density.
Q2	Link prediction over the typed SDLC graph (GNN / KG-embedding) adds signal beyond pairwise text cosine.	**Partly**: a parameter-free lift helps issue→PR; a learned R-GCN does not beat it at pilot scale.
Q3	Relations with weaker surface text (diff→test, log→file) show a larger relational lift than issue→PR.	**Open / blocked** by co-change density (structure-bound at pilot).
Q4	Cross-repo same-bug-class retrieval benefits from relation-trained embeddings (a latent, non-hyperlinked relation).	**Not yet run** (Track C2).
Q5	Scale (more repos / pairs) gives a learned head the headroom it lacked at pilot scale.	**Open**: data scales; learned-head re-test is a GPU follow-up.
Q6	A code-specific embedder beats a strong general one (how much does the base matter?).	**Refuted as stated**: the axis is *embedding-tuned*, not 'code'.

Table 4. Open questions and current verdicts. A 'no' reroutes the program (diagnose / change base / scale); it does not stall it.

6 Scope of Experiments

Experiments run as **waves** (R3–R16E), each a single change against the appropriate control, organized into parallel tracks with explicit decision gates. A gate outcome — win or honest negative — routes the next wave; scale is spent only on a method that has already shown signal.

Track	Theme	State
A — Representation	Fine-tune the encoder with the relation loss (LoRA).	**Won** (pilot → dense Tier-2); the program's core positive result.
B — Graph	Link prediction / aggregation over the SDLC graph.	Probe done; free-aggregation lift robust; learned GNN re-gated on scale.
C — Tasks and labels	diff→test, log→file; cross-repo same-bug-class.	diff→test characterized (structure-bound); same-bug-class pending.
D — Scale (on signal only)	Multi-split CIs; more/denser repos.	5-split confidence done; 78-repo dense Tier-2 done; 200–500 repos is GPU-scale.

Track	Theme	State
E — Beyond MVP-1	Relational SLM, GraphRAG, agent policy.	Entry started (subgraph packer + SLM dry-run); gated behind a solid retrieval win.

Table 5. The five tracks and their gate state.

6.1 Method invariants and the audit

Six invariants make results comparable: links-as-labels; frozen cross-repo splits as the headline; pretrained embeddings as the substrate with the relational contribution measured as the delta over embedder-cosine; an experiment card per run; reproducibility from a clean checkout on numpy; and honest negatives kept. An **independent adversarial audit (R13)** stress-tested the five headline claims and returned **sound, zero CRITICAL issues**: cross-repo splits are genuinely disjoint, the reference scrub leaks no gold-pair numbers, the graph leakage guard is load-bearing, and every numpy result reproduces the committed cards byte-for-byte. A CI provenance test now guards disjointness mechanically.

7 Results

7.1 The cross-repo substrate is settled, and the head fails

System (de-referenced, cross-repo; 174 held-out queries)	R@1	R@5	MRR
IDF cosine (bag-of-tokens bar)	0.460	0.828	0.624
embedder-cosine (frozen MiniLM-L6)	**0.592**	0.920	**0.728**
embedder + from-scratch head	0.190	0.644	0.381
embedder + identity-init operator	0.592	0.931	0.737

Table 6. Pretrained embeddings win cross-repo with no training (0.46 → 0.592). A from-scratch head on frozen vectors is actively harmful (0.190); an identity-init operator is safe but adds nothing. The relational contribution cannot be a post-hoc head on frozen features. (Earlier rungs: bag-of-tokens projections fail cross-repo — two-tower 0.241 < vanilla 0.391 — because tokens do not transfer between repos; only meaning does.)

7.2 The relational fine-tune wins, and the win grows with data

LoRA vs. frozen, cross-repo	Frozen R@1	LoRA R@1	ΔR@1	ΔMRR	Held-out repos
pilot (20 repos, 182 train pairs)	0.592	**0.655**	+0.07	+0.063	8
scale (55 repos)	—	—	+0.080	+0.072	14
dense Tier-2 (78 repos, ~35 q/repo)	0.515	**0.629**	**+0.114**	+0.101	32
5-split confidence (pilot re-splits)	—	—	+0.061±0.021	+0.052±0.010	5 splits

Table 7. A 0.48%-parameter adapter trained on CPU generalizes to unseen repositories, and a head on the tuned vectors still adds nothing (the gain is in the representation). The progression is monotone in density (pilot +0.07 → scale +0.080 → Tier-2 +0.114): more positive pairs per repo give the relation loss more to contrast. A sweep (R16D) shows +0.114 is near-tuned, not a floor — rank is saturated (r8 ≈ r16 ≈ r32, ±0.003) and harder in-batch negatives are the only remaining lever (best r16-b48: +0.126), now bounded by CPU memory.

7.3 A thin graph lift stacks on top — robust, and structure-bound where it fails

Graph lift (typed mean-aggregation)	R@1	Note
issue→PR, frozen + graph	0.621	vs. frozen-cosine 0.592
issue→PR, LoRA + graph	**0.690**	vs. LoRA-cosine 0.655; best at hops=1, α≈0.25
learned 1-hop R-GCN (issue→PR)	0.575	overfits ~180 pairs; < free-agg 0.609
diff→affected-test (any α, any hops)	0.009	flat: 46.9% of gold tests isolated → 59.8% reachable ceiling

*Table 8. The R16E α×hops sweep shows the issue→PR lift is a **plateau** (positive across α in [0, 0.75] on both frozen and LoRA features) and **saturates at 1 hop** (hops=1 ≡ hops=2, byte-for-byte) — so R11B's 0.690 is robust and cheap, not a tuned spike, and α=1.0 reproduces embedder-cosine exactly (sanity anchor). diff→test is flat at every setting because, once the gold edge is*

honestly removed, ~47% of positive test nodes are degree-0 — a feature- and hop-independent ceiling. The limiter is co-change density (a data problem), not the method.

7.4 Two counter-intuitive negatives, and a chunking mirror

- **Base model: 'embedding-tuned', not 'code'.** A code-MLM (codebert) collapses to R@1 0.144; the axis is monotone in embedding-tuning — codebert 0.14 < unixcoder 0.45 < st-codesearch 0.55 < MiniLM 0.592 ≈ bge 0.598. A true code-*embedding* base (jina-code) gets the best R@5 ever (0.960) and ties MRR but does not beat top-1.
- **More text hurts.** A paired control de-truncating bodies (500 → 8000 chars) *lowers* every system by 0.09–0.15 R@1 (embedder 0.69 → 0.55): the first ~500 chars carry the signal and the rest dilutes it. Truncation was a feature.
- **Chunking confirms the mechanism.** For front-loaded issue→PR, **FirstP** (the lede) wins at every chunk size (FirstP@512 0.701 > MaxP 0.668). The mirror image: for deep-signal diff→affected-test, **MaxP** wins at every chunk size (ΔR@1 up to +0.346 at small chunks), exactly as predicted.
- **Bag-of-tokens ordering is stable at scale.** At 55 repos IDF 0.444 ≥ vanilla 0.333, and at dense Tier-2 IDF 0.389 vs. vanilla 0.287 (+0.102) — the frequency signal generalizes.

7.5 Full results ledger

Wave	Hypothesis (abbrev.)	Finding	Headline
Synthetic per-token	relation supervision > vanilla/IDF when the link is per-token	**confirmed**	R@1 0.82 vs IDF 0.38 vs van. 0.11
Cross-token synthetic	only a cross-token operator bridges disjoint vocab	**confirmed**	tower R@1 0.83 vs all-else 0.01
Real issue→PR (linked)	does the synthetic win transfer?	**no** — surface-rich; IDF ties diagonal	IDF R@1 0.46
De-ref. cross-repo (tokens)	learned head generalizes cross-repo?	**no** — tower 0.24 < vanilla 0.39	tower 0.24
Embeddings cross-repo	pretrained embeddings generalize?	**yes** — embedder-cosine wins	R@1 0.59 vs IDF 0.46
Head on frozen embeddings	does our operator add value on top?	**no** at pilot — from-scratch 0.19; identity ties 0.59	—
LoRA fine-tune (A)	relation loss inside the rep beats frozen?	**YES** (pilot)	R@1 0.66 vs 0.59; MRR 0.79
Multi-split (R11A, D)	is the LoRA win robust, not split luck?	**YES** — positive on all 5 splits	ΔR@1 +0.061±0.021
Graph probe (R11B, B)	does typed aggregation add beyond cosine?	**small but real; stacks with LoRA**	LoRA+graph 0.69
Base model (R12B, Q6)	does a code/stronger base help?	**embedding-tuned matters, not 'code'**	bge 0.598 / MiniLM 0.592 / codebert 0.144
Learned R-GCN (R12B)	does a learned GNN beat free aggregation?	**no** at pilot — overfits ~180 pairs	rgcn 0.575 < free-agg 0.609
Scale ~55 repos (R12A)	does bag-of-tokens hold at scale?	**yes** — IDF still best	IDF 0.444 ≥ van. 0.333
Relational SLM v0 (R12C, E)	can a subgraph drive an SLM (MVP-2)?	**dry-run runs** (no benchmark claim)	retrieval ~0.9 top-5
LoRA-at-scale (R13A, D)	does the LoRA win hold on 55 repos?	**yes — and grows**	ΔR@1 +0.080 (14 test repos)
Code base (R13B, Q6)	does a code base beat the general substrate?	**no** — monotone in embedding-tuning	codebert 0.14 ... MiniLM 0.59 ≈ bge 0.60
Full text (R14)	does de-truncating bodies help?	**no — it HURTS** (paired control)	−0.09 to −0.15 R@1; embedder 0.69→0.55
Chunked MaxP (R15)	does MaxP beat FirstP for issue→PR?	**no** — signal front-loaded	FirstP@512 0.701 > MaxP 0.668
Code base pinned (R15B)	does a true code+embedding base win?	**qualified yes** — best R@5, ties MRR	R@5 0.960; R@1 0.580
Deep-content chunk (R16A)	does MaxP win where signal is deep?	**yes** — mirror of R15	ΔR@1 up to +0.346
Dense Tier-2 base (R16B, D)	does bag-of-tokens hold at 78 dense repos?	**yes** — density did not erase the gap	IDF 0.389 vs van. 0.287
LoRA-at-Tier-2 (R16C, D)	does the LoRA win hold at dense ~80 repos?	**yes — and grows further**	ΔR@1 +0.114 (0.515→0.629)
LoRA sweep (R16D, D)	is +0.114 a floor or near-tuned?	**near-tuned** — rank saturated	best r16-b48 ΔR@1 +0.126
Graph-lift sweep (R16E, B)	tuned knife-edge or robust; can hops rescue diff→test?	**robust plateau + structure-bound**	issue→PR 0.690 (h1); diff→test ceiling 59.8%
LoRA-win CIs (R17a, A/D)	does the headline delta survive a within-split CI?	**yes** — both query- and repo-cluster 95% CIs exclude zero; broad but not uniform (5/8 repos)	ΔR@1 +0.063, CI [+0.006,+0.121]; ΔMRR CI [+0.027,+0.102]
diff→test density (R17b, B)	is the 59.8% ceiling method-bound or an ingest-depth artefact?	**density artefact** — gold tests heavily co-changed (median 35 commits)	reachable ceiling 59.8% → 96.4% under real co-change

Wave	Hypothesis (abbrev.)	Finding	Headline
Negatives lever (R18, A/D)	more vs harder in-batch negatives on the M5 GPU?	**harder, not more** — random pool flat (b192/384 OOM the M5); same-repo hard batching is the new best	repo-hard ΔR@1 +0.137, CI [+0.112,+0.164], 31/32 repos up
Hardness multi-seed (R19, A/D)	is the hardness gain real across seeds or MPS noise?	**real** — paired gap positive on both seeds, tight (mean +0.020, std ~0.001)	random +0.122 vs repo-hard +0.142; mining (H2) deferred

Table 9. Full results ledger (waves R3–R16E). The committed experiment cards under data/cards/examples/ are authoritative for every number.

8 Conclusions

The relational win lives in the base representation. Across every experiment the single lever that consistently pays off is *changing the representation itself* — an embedding-tuned substrate, LoRA reshaping it with the relation loss, and a thin parameter-free graph lift on top. The single thing that consistently fails is *adding a learned head over frozen features* — a from-scratch tower, an identity-init operator, a learned R-GCN all overfit at pilot scale and do not help.

What is settled at pilot/Tier-2 scale:

- **Pretrained embeddings are the cross-repo substrate.** Meaning transfers across repositories where tokens do not; a frozen semantic embedder beats every bag-of-tokens system on held-out repos with zero training.
- **The contribution belongs in the representation** (LoRA), not a post-hoc head — and the win is robust (all 5 splits) and grows monotonically with per-repo density (+0.07 → +0.114).
- **The graph lift is real but thin and 1-hop** on issue→PR; diff→test's 59.8% ceiling is an **ingest-depth artefact** — real co-change history lifts it to 96.4% (R17b), so the blocker is removable by denser data, not the method.
- **Base = embedding-tuned, not 'code'**; and **more text is not better** — both counter-intuitive, both from paired controls.

What remains open is honest and bounded: the learned-head and learned-GNN re-tests at GPU scale; whether weaker-surface relations (diff→test, log→file) beat issue→PR once the data is dense enough; cross-repo same-bug-class retrieval; and the remaining paper losses (L_contrastive, L_rel, L_graph, L_logic, L_align). The lab's enduring deliverable is the frozen public benchmark with honest baselines and full provenance that makes exactly these questions measurable.

9 Next Steps

Direction	Concrete next action	Gate / prerequisite
Scale the LoRA win	Tier-2 full breadth (200–500 repos); harder/larger in-batch negative pools (the one lever R16D left).	GPU memory (CPU-bound today).
Learned graph model	Richer/regularized multi-hop R-GCN on pretrained node features; KG-embedding scoring (DistMult/ComplEx/RotatE).	Track-D scale for supervision; must beat ~0.690 (LoRA) / ~0.621 (frozen), not raw cosine.
Unblock diff→test	R17b showed the 59.8% ceiling is an ingest artefact (real ceiling 96.4%); next is the retrieval re-eval on a denser graph with embedded PR nodes.	Torch to embed new nodes — the ceiling no longer blocks it.
New tasks/labels	Activate log→file; cross-repo same-bug-class via a curated SWE bug-class ontology (labels independent of body text).	Non-degenerate (not regex-recoverable) before it counts.
Long-text handling	Salient-section selection / better pooling — not naive whole-body (which hurts).	Beat the FirstP lede baseline.

Direction	Concrete next action	Gate / prerequisite
Relational SLM (MVP-2+)	QLoRA SLM with relation/policy heads (review, test selection, risk) + GraphRAG packer; outcome learning.	Gated behind a solid retrieval win — which Track A now provides.

Table 10. The decision-gated next steps. Per the program's compute discipline, GPU and broad scale are spent only where a method has already shown signal.

All results are exploratory and pilot/Tier-2 scale; they are signals to confirm, not release-quality evidence. Where this report and the repository's committed experiment cards disagree, the cards are authoritative.